Surgical Urgency and Perioperative Mortality: A National Analysis Using Procedure-specific Classification

To the Editor:

Recent work in Anesthesiology reported perioperative mortality of 0.3% among electively and 2.3% among nonelectively hospitalized Florida inpatients—a 7.7-fold difference.[1] The authors concluded that surgical urgency is the primary mortality determinant and recommended broad urgency-based quality improvement over procedure-specific interventions. However, the methodology used did not permit evaluation by procedure type,[2] and aggregate rates have limited utility for prognostication; it cannot be assumed that every elective procedure carries a 0.3% risk or every nonelective procedure a 2.3% risk. The relationship between urgency and risk is more nuanced, and effective mitigation requires identification of the highest-risk groups and modifiable contributors. To address these gaps, this analysis examined a nationally representative cohort to validate previous mortality findings,[1] compare procedure-specific mortality between elective and nonelective hospitalizations, and identify diagnoses and conditions derived from an observational dataset that are associated with high-mortality nonelective populations.

This retrospective cohort study used the 2022 Healthcare Cost and Utilization Project National Inpatient Sample, a 20% stratified sample of U.S. hospital discharges that covers 97% of the U.S. population.[3] It includes information on patient demographics, International Classification of Diseases, Tenth Revision (ICD-10) diagnosis and procedure codes recorded as part of the hospitalization, procedure timing, length of stay, and in-hospital mortality—data typically available from discharge abstracts. The National Inpatient Sample enables weighted national estimates of healthcare utilization and outcomes.

Patients aged 18 yr or older undergoing major surgical procedures were included. Elective status was defined as elective admission, surgery performed on admission day, and absence of emergency department codes. Records missing key variables were excluded: timing of first procedure (1.7% missing), length of stay (0.005% missing), and mortality status (0.014% missing).

First, to validate previous state-level findings by Epstein *et al.*, procedures were classified using the Procedure Classes Refined system, which categorizes major surgical procedures into diagnostic or therapeutic groups.[1] For more granular comparisons, the U.S. Centers for Disease Control and Prevention (Atlanta, Georgia) National Healthcare Safety Network Operative Procedure Classification was then applied, which categorizes procedure types into 39 categories by anatomic site and surgical approach, focusing on operations that require operating room resources.[4,5] This approach enabled comparison of identical procedures across urgency levels, unlike broader classification systems that group heterogeneous procedures together. The primary outcome was death at any time during hospitalization.

Weighted national estimates of procedure volumes and mortality rates with 95% CIs were generated, accounting for the National Inpatient Sample sampling design. Chi-square tests with Bonferroni correction were used for multiple comparisons (20 in total). Proportional mortality contribution analysis compared each procedure's share of deaths with its share of procedures, highlighting categories disproportionately contributing to mortality.[6] Enrichment analysis, which identifies whether specific outcomes or characteristics occur more frequently in a subgroup than expected by chance, captured diagnosis and procedure codes disproportionately represented in nonelective (*versus* elective) cases among the five highest-risk (observed mortality) procedural categories, quantified using risk ratios.[7,8] Figures 1 and 2 display the 10 most frequent nonelective procedure categories.

Analyses were performed using SAS version 9.4 (SAS Institute Inc., USA). The full code is publicly available (https://github.com/ohyder/NHSN_Mortality).

In the 2022 National Inpatient Sample cohort, mortality was 0.3% for elective and 2.4% for nonelective hospitalizations in both the Procedure Classes Refined (7,842,270 hospitalizations) and U.S. Centers for Disease Control and Prevention National Healthcare Safety Network classifications (5,019,561 hospitalizations), similar to previous state-level findings (Supplemental Digital Content table S1, https://links.lww.com/ALN/E396).

Procedure-specific analysis demonstrated marked heterogeneity (fig. 1). Abdominal aortic aneurysm repair showed the largest absolute mortality difference between elective and nonelective cases (3.4% *vs.* 18.7%), followed by operations on the spleen (1.6% *vs.* 12.7%) and neurosurgical

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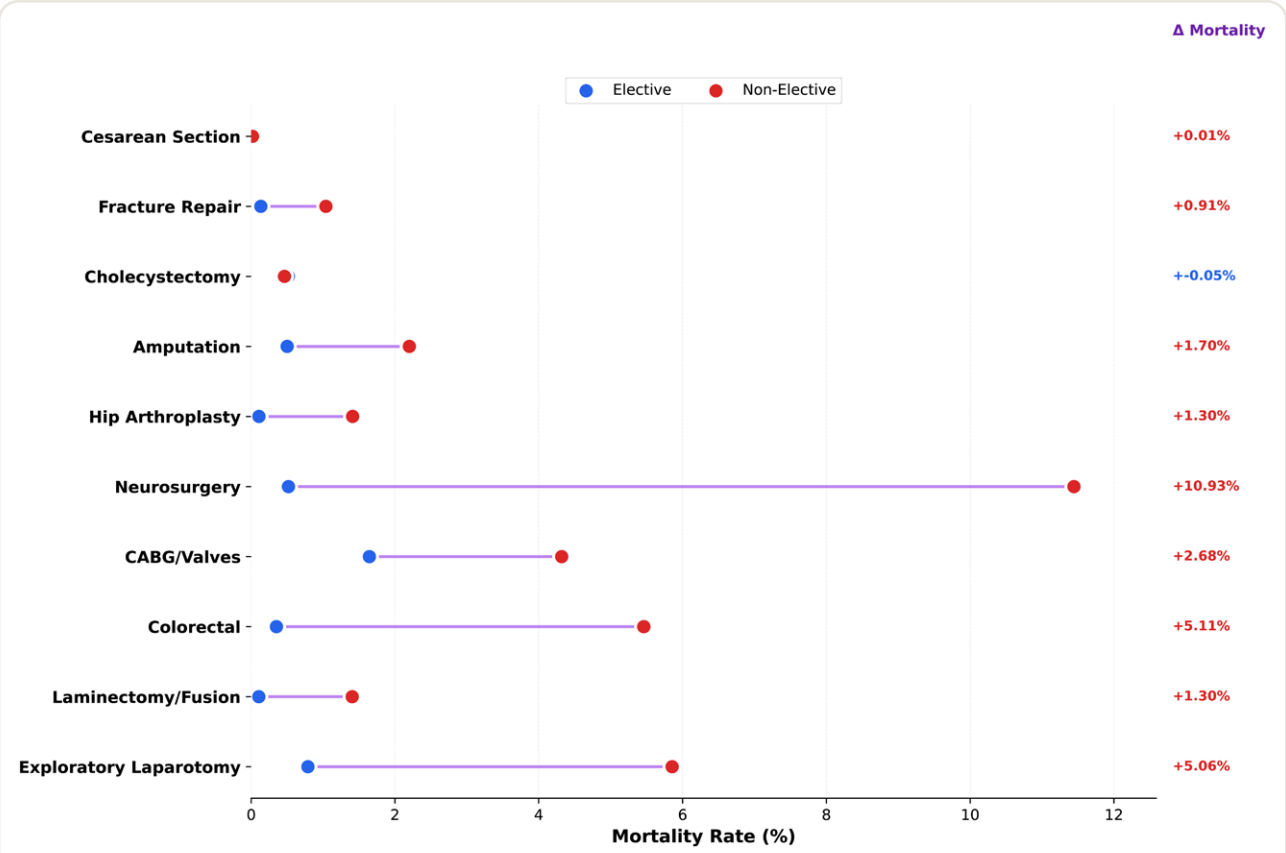

Fig. 1. Cleveland dot plot showing heterogeneity in procedure-specific mortality among elective *versus* nonelective cases for the 10 most common nonelective U.S. Centers for Disease Control and Prevention (Atlanta, Georgia) National Healthcare Safety Network categories by national procedure volume. Δ Mortality is the difference in mortality for each category when comparing nonelective and elective hospitalizations. CABG, coronary artery bypass graft.

procedures (0.5% *vs.* 11.4%) (all $P < 0.001$), representing 5- to 21-fold differences. In contrast, cesarean delivery mortality was less than 0.1% regardless of urgency.

Discrepancies were present between each procedure's share of total procedures and its share of total deaths. Coronary artery bypass graft (CABG) and cardiac valve procedures accounted for 5.5% of elective cases but 31.5% of elective deaths (+26.0 percentage points). Neurosurgical procedures represented 5.5% of nonelective cases but 26.1% of nonelective deaths (+20.6 percentage points). Additional procedures overrepresented in nonelective mortality included colorectal (5.1% of cases, 11.6% of deaths), small bowel (2.9% of cases, 8.9% of deaths), exploratory laparotomy (3.6% of cases, 8.9% of deaths), and CABG/valve (5.3% of cases, 9.6% of deaths) cases (all $P < 0.001$). Figure 2 contrasts each procedure's procedural share with its mortality share.

Enrichment analysis showed that nonelective neurosurgery was associated with diagnoses of traumatic brain compression, subdural hemorrhage, and ischemic stroke due to large-vessel occlusion. Corresponding interventions included clot retrieval from intracranial vessels and subdural hematoma evacuation. Colorectal procedures were associated with diagnoses of rectal abscess requiring drainage, peritonitis, and septic shock. Small-bowel surgery was associated with bowel obstruction, perforation, and sepsis. Nonelective cardiac surgery was linked to acute myocardial infarction, with procedures including insertion or removal of circulatory support devices. Detailed enrichment analysis results, including specific ICD-10 codes, risk ratios, and prevalence rates, are provided in Supplemental Digital Content table S2 (https://links.lww.com/ALN/E397).

This national analysis validated previous state-level findings[1] and demonstrated that mortality was not uniform within urgency categories but concentrated in specific high-risk procedures. Enrichment analysis provided insight into this concentration, identifying organ ischemia, trauma, and infectious processes as diagnoses associated with nonelective procedures with disproportionately high mortality. Proportional mortality analysis showed that many of the common nonelective procedures—cesarean delivery, appendectomy, and cholecystectomy—had minimal mortality, while almost a third of elective deaths were concentrated in CABG and heart valve surgery. These findings suggest

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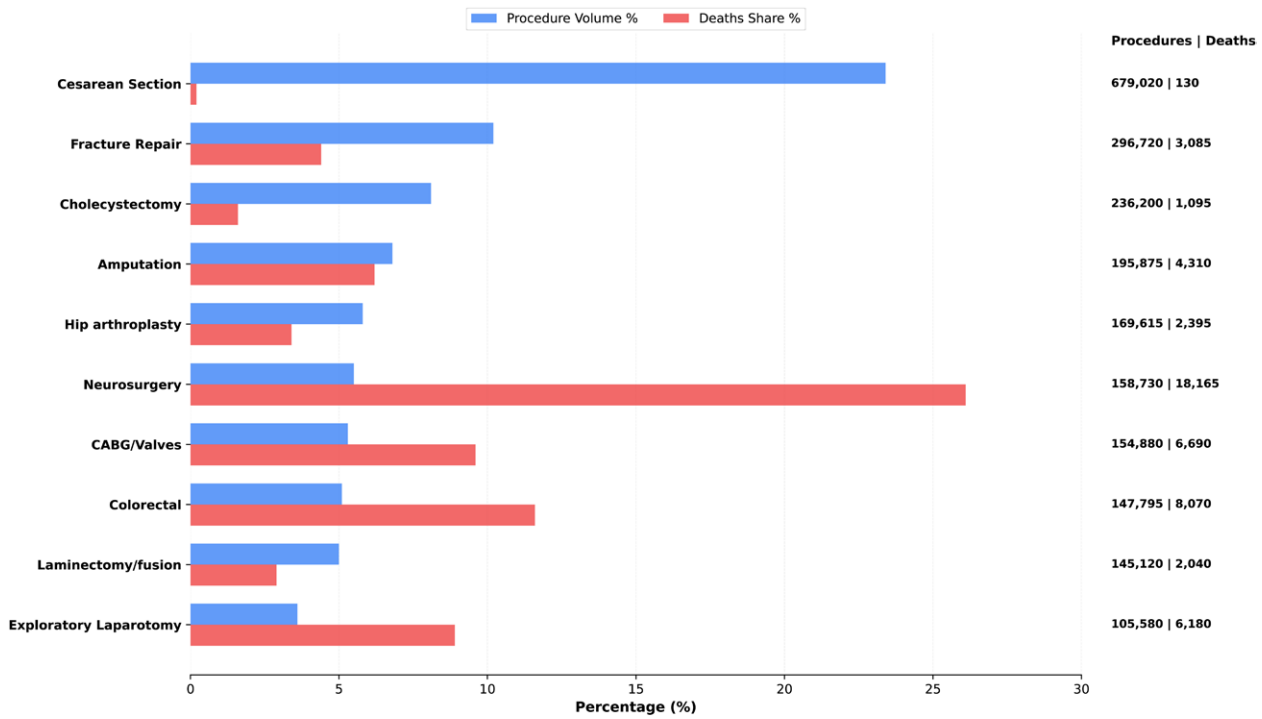

Fig. 2. Distribution of procedure volume *versus* mortality for the 10 most common nonelective U.S. Centers for Disease Control and Prevention (Atlanta, Georgia) National Healthcare Safety Network categories by national procedure volume. *Blue bars* show each procedure/category as a percentage of all procedures. *Red bars* show each procedure's deaths as a percentage of all deaths. When the *red bar* is higher than the *blue bar*, that procedure is overrepresented. The *right column* shows the number of hospitalizations and deaths within the category. CABG, coronary artery bypass graft.

that transitioning to broad urgency-based approaches to reduce perioperative mortality, without accounting for procedure-specific risk patterns, would be insufficient. Instead, targeted interventions for high-risk surgical patients and procedure-specific quality improvement efforts should continue as our understanding of urgency–risk relationships evolves, with relevance for perioperative decision-making. The data support integrating procedure-specific risk profiles into urgency-based perioperative pathways.

Strengths of this analysis include the use of a large, nationally representative cohort and standardized classification of major surgical procedures, supporting generalizability to U.S. hospitals. Additionally, the enrichment analysis methodology systematically evaluated thousands of ICD-10 codes to identify hypothesis-generating patterns for future outcomes research. Key limitations of this analysis were the lack of clinical granularity and temporal certainty for diagnoses in the 2022 Healthcare Cost and Utilization Project National Inpatient Sample, which prevented detailed assessment of indications, comorbidity burden, disease severity, complications, and hospital-level factors.

Competing Interests

The authors declare no competing interests.

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Supplemental Digital Content

Supplemental Table S1. Characteristics and outcomes (National Inpatient Sample 2022), https://links.lww.com/ALN/E396

Supplemental Table S2. Enriched diagnoses and procedures, https://links.lww.com/ALN/E397

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